

GRAND STRAND, SC, USA

WMO: 747915

Lat: 33.816N

Lon: 78.720W

Elev: 32

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 93718

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.3	28.2	7.1	7.9	32.0	12.0	10.1	35.7	22.2	59.8	19.4	56.9	4.4	20	0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	90.4	78.2	88.5	78.0	87.1	77.5	81.2	86.7	80.3	85.8	79.3	84.9	10.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.5	153.0	84.6	78.7	149.1	84.2	77.4	142.8	83.4	44.7	86.9	43.7	85.7	42.7	84.8	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.4	18.3	16.6	DB	19.0	95.3	3.2	4.0	16.7	98.2	14.9	100.5	13.1	102.7	10.8	105.6
			WB	16.7	82.6	2.9	1.0	14.6	83.3	12.9	83.9	11.2	84.4	9.1	85.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.1	45.9	48.6	54.7	62.7	70.8	77.9	80.2	79.8	75.6	66.1	56.0	50.0
	DBStd	14.02	9.71	8.03	8.11	6.71	5.65	4.02	3.23	3.23	4.52	7.35	8.16	9.02
	HDD50	516	198	114	49	3	0	0	0	0	0	2	33	117
	HDD65	2351	591	459	329	119	18	0	0	0	2	78	285	470
	CDD50	5666	72	76	193	383	644	838	936	923	769	501	214	117
	CDD65	2024	0	1	8	49	197	388	471	458	321	112	15	4
	CDH74	17848	4	1	24	141	1109	3642	4967	4682	2584	661	32	1
	CDH80	5118	1	0	3	23	167	970	1672	1561	647	73	1	0
(14) Wind	WSAvg	7.5	6.8	7.4	8.1	8.7	8.5	8.4	8.1	7.2	7.3	6.4	6.2	6.4
(15) Precipitation	PrecAvg	51.0	3.5	3.5	3.8	2.8	3.7	4.8	5.8	6.7	6.3	4.1	2.8	3.3
	PrecMax	70.1	8.8	12.0	8.1	6.5	7.2	12.7	10.4	14.2	21.4	13.5	5.9	10.1
	PrecMin	34.3	0.8	0.9	1.1	0.1	1.3	0.4	3.0	3.3	1.6	0.1	0.2	1.0
	PrecStd	8.3	2.0	2.1	1.8	1.7	1.5	2.6	1.9	2.5	4.4	3.3	1.5	2.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.4	72.2	76.8	83.1	89.1	93.0	93.8	92.7	89.1	84.9	78.4	72.7
		MCWB	62.1	61.1	62.2	65.9	74.3	77.1	78.9	80.0	75.5	74.2	69.9	66.4
	2%	DB	66.2	67.3	72.5	77.2	83.8	89.2	90.5	89.7	86.6	81.8	74.1	69.8
		MCWB	60.7	58.5	62.5	67.1	72.6	77.1	78.9	79.2	75.6	72.7	66.4	64.7
	5%	DB	63.6	64.3	69.8	74.8	81.3	86.6	88.4	87.9	85.1	80.0	72.2	67.0
		MCWB	59.4	58.3	61.7	66.3	72.4	76.7	78.7	78.5	75.4	71.4	65.5	63.1
	10%	DB	61.1	61.6	66.5	72.8	79.4	84.5	86.9	86.4	83.7	78.1	69.9	64.3
		MCWB	57.3	57.1	60.2	65.4	71.5	75.9	78.1	77.8	74.9	70.5	64.2	60.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.1	65.9	67.4	72.9	77.4	81.0	82.4	82.8	80.2	77.6	72.2	69.1
		MCDB	67.8	69.4	71.4	76.4	83.8	86.8	88.0	88.8	83.8	81.0	75.3	70.9
	2%	WB	62.7	62.5	65.3	70.3	75.5	79.7	81.3	81.4	78.6	75.7	69.6	66.1
		MCDB	65.1	64.9	69.8	74.8	80.7	85.5	86.9	86.8	83.4	79.8	72.1	68.2
	5%	WB	60.5	60.2	63.6	68.6	74.2	78.4	80.3	80.3	77.7	74.1	67.2	63.8
		MCDB	63.1	63.0	67.2	72.8	79.2	84.1	86.1	85.7	83.1	78.4	70.4	66.0
	10%	WB	58.0	57.7	62.0	67.1	73.1	77.5	79.3	79.2	76.5	72.3	65.2	61.8
		MCDB	60.6	60.8	65.6	71.2	77.9	83.1	85.2	84.7	81.9	76.6	68.6	64.2
(35) Mean Daily Temperature Range	5% DB	MDBR	17.9	18.0	18.1	16.9	14.9	13.2	12.8	12.6	14.3	17.6	19.6	18.1
		MCDBR	18.8	19.0	18.8	17.0	15.3	14.5	14.4	13.8	14.9	17.1	19.5	17.8
		MCWBR	14.8	13.1	11.1	9.9	7.7	7.2	7.2	6.6	7.0	9.4	12.9	13.8
	5% WB	MCDBR	16.5	16.5	16.0	13.4	12.5	12.5	13.1	12.2	12.0	12.7	15.4	16.1
		MCWBR	14.6	13.7	11.3	8.7	7.0	7.0	7.5	6.7	6.7	8.1	12.2	14.0
(40) Clear-Sky Solar Irradiance	taub		0.327	0.335	0.367	0.398	0.458	0.484	0.525	0.502	0.429	0.393	0.352	0.331
	taud		2.518	2.492	2.427	2.324	2.211	2.185	2.086	2.148	2.344	2.414	2.485	2.537
	Ebn at Noon		281	291	288	282	265	257	246	250	265	266	269	273
	Edh at Noon		27	30	35	40	46	47	51	48	37	32	27	25
(44) All-Sky Solar Radiation	RadAvg		866	1079	1444	1835	1999	2063	1958	1760	1497	1256	963	770
	RadStd		54	106	128	122	135	128	122	99	111	134	72	59

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+1.42	N/A	N/A	N/A	N/A	N/A	N/A	+434	+288
(47) Regional (48 neighbors)		+0.85	N/A	N/A	N/A	N/A	N/A	N/A	+293	+212

Nomenclature: See separate page